KMOU Division of Marine System Engineering

Introduction to Internal Combustion Engine

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Lesson content for each week

Ch0_Introduction to the Class

Ch1_Overview of the Internal Combustion Engine

Ch2_Thermodynamics and Compressed Air System

Ch3_Marine Fuels and Internal Fuel Oil System

Ch4_Intake and Exhaust System

Ch5_Turbocharger

Midterm

Ch6_Piston Fittings and Crankshaft

Ch7_Lube Oil System

Ch8_Cooling Water System

Ch9_Introduction to the Main Engine

Ch10_Alternative Fuels and Alternative Fuel Systems

Ch11_Environmental Regulations and Exhaust Gas Aftertreatment Systems

Ch12_Tools, Apprentice Engineer Duties, and Career Direction

Final Exam

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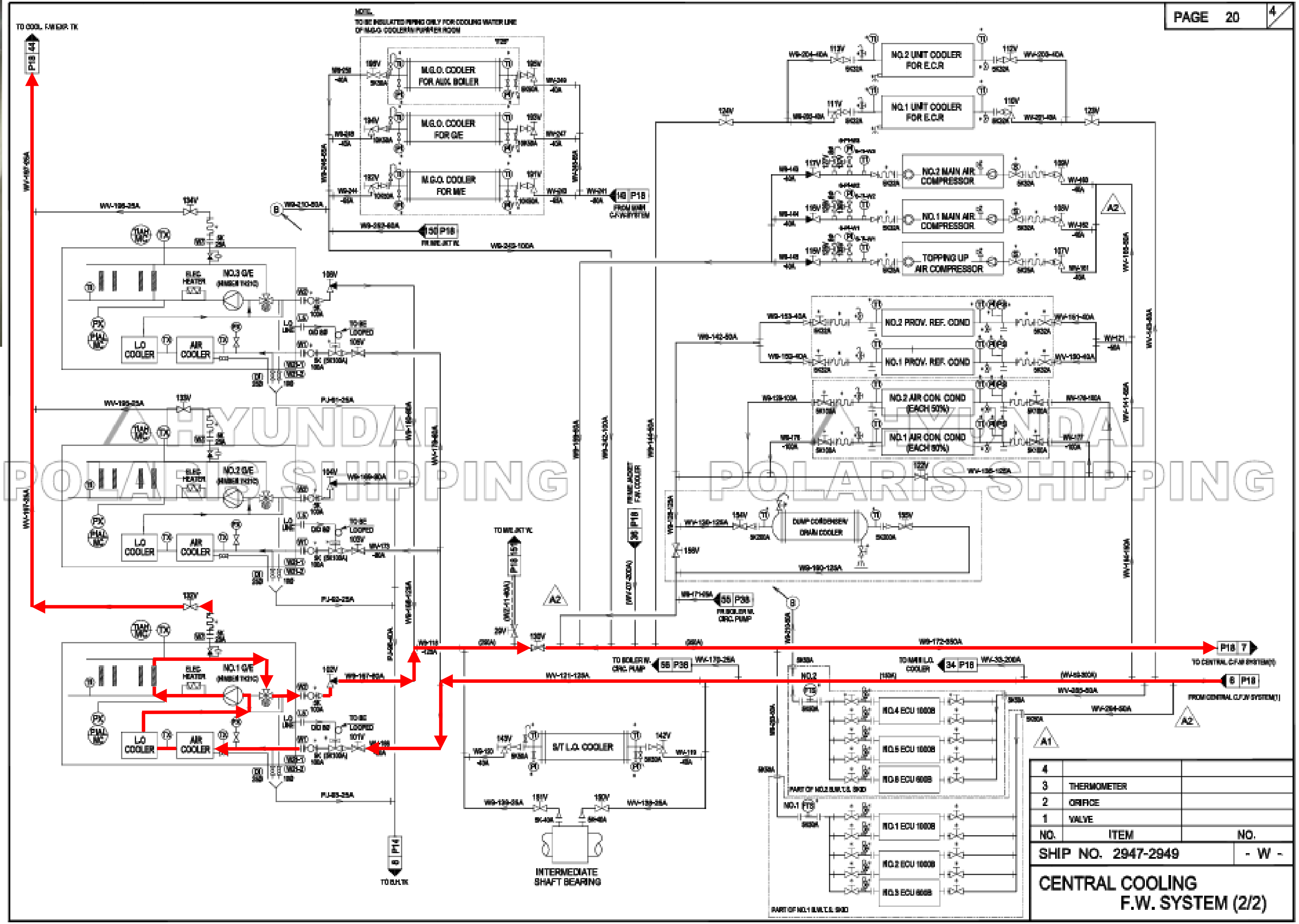
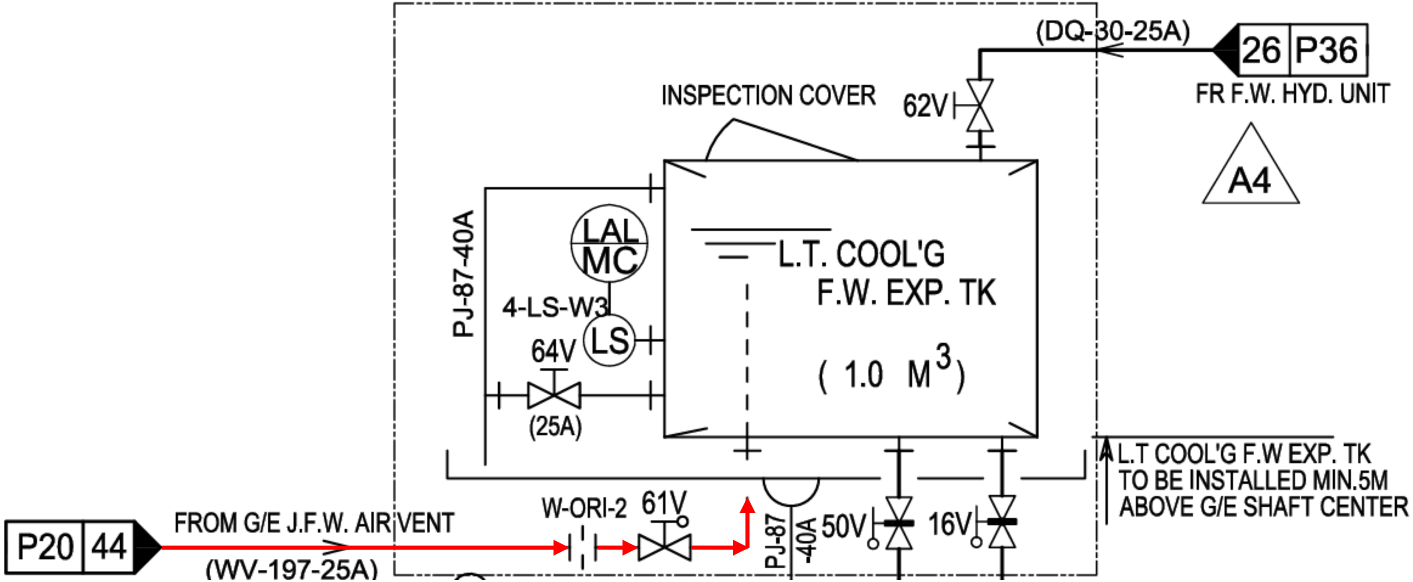
Cooling fresh water expansion tank



Cooling fresh water expansion tank



Cooling fresh water expansion tank

Cooling water storage and replenishment function

Stores cooling water (fresh water) used to cool the engine. It replenishes cooling water lost to evaporation, leakage, etc. during operation.

Provides space for cooling water expansion (acts as an expansion tank)

When the engine is running, the cooling water heats and expands. The cooling fresh water expansion tank receives this expanded cooling water, maintaining stable cooling system pressure.

Cooling water degassing

During operation, air (gas) bubbles may form in the cooling water. The tank provides a space for this gas to separate and escape, preventing cavitation and ensuring circulation stability.

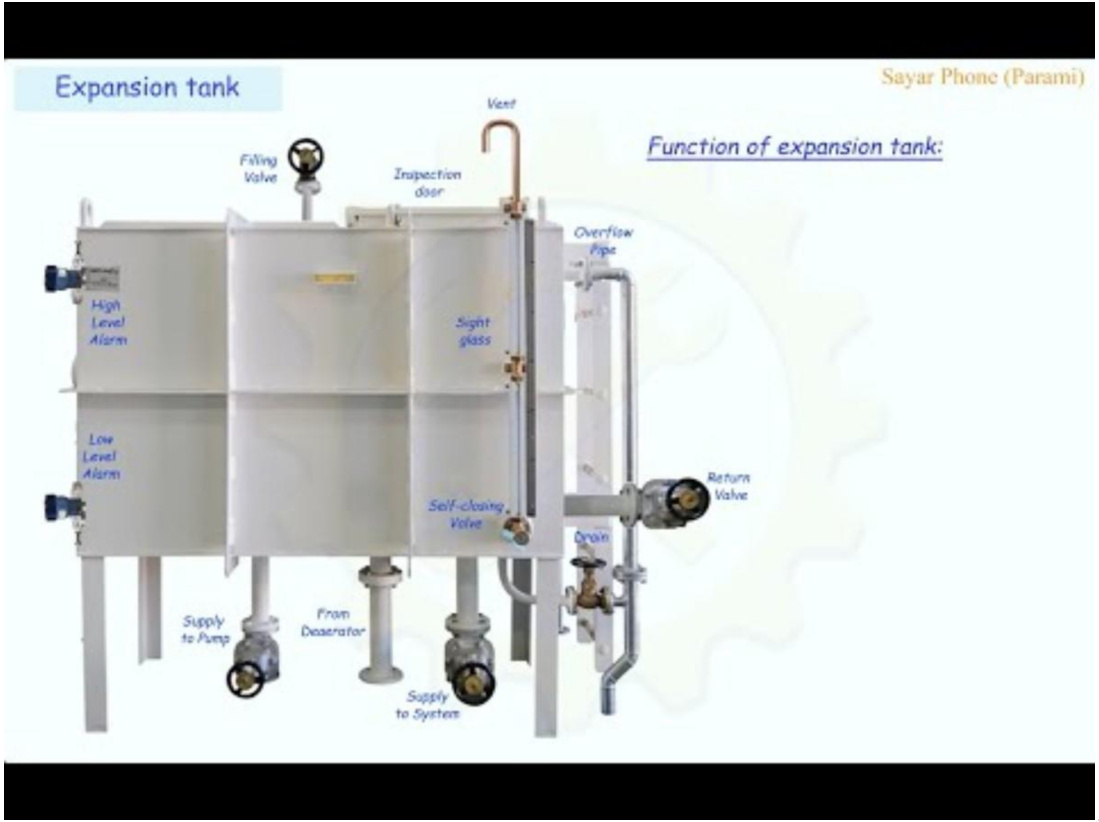
Maintains circulation pressure

Since the cooling water system is a closed circulation system, the tank acts as a head tank, maintaining a constant water head. This ensures stable inlet pressure to the pump.

Cooling water quality control

Cooling fresh water expansion tanks typically have chemical inlets, allowing the addition of corrosion inhibitors or alkalinity regulators to prevent corrosion and scale.

Cooling fresh water expansion tank



Cooling fresh water expansion tank

Cooling water system (HiMSEN)

The engine has two cooling water circuits internally, which are low temperature (LT) system and high temperature (HT) system. Most of the elements of the circuits are modularized and directly mounted on the feed block.

The cooling water system is designed for using normal fresh water with corrosion inhibitor. Both the low and the high temperature system are cooled by fresh water.

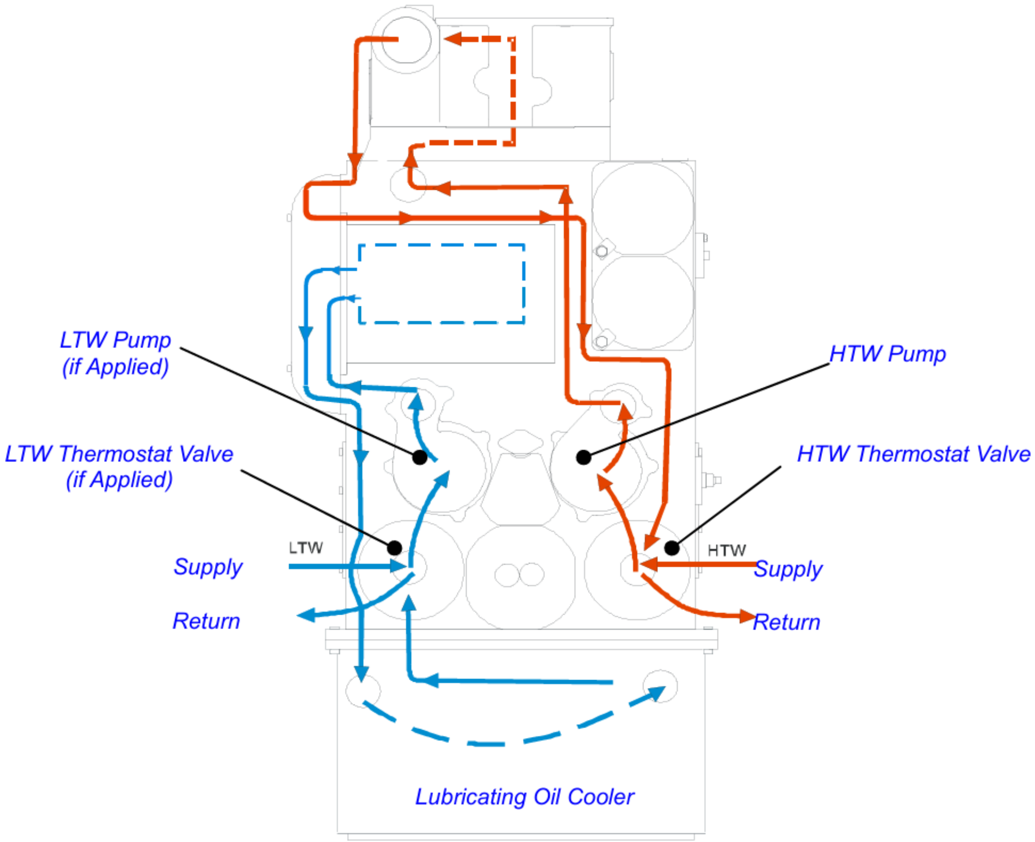


Fig.1 Circuit diagram for internal cooling water system (내부 냉각수 계통도)

Cooling water system (HiMSEN)

The low temperature cooling water circulates by L.T pump and flows through charge air cooler and lubricating oil cooler internally.

L.T.W thermostat valve controls the water temperature by exchanging water with external circuit as shown in fig. 1.

The high temperature cooling water circulates by H.T pump internally and flows through cylinder head and water jacket (for cylinder liner) under normal operation. H.T.W thermostat valve controls the water temperature by exchanging water with external circuit.

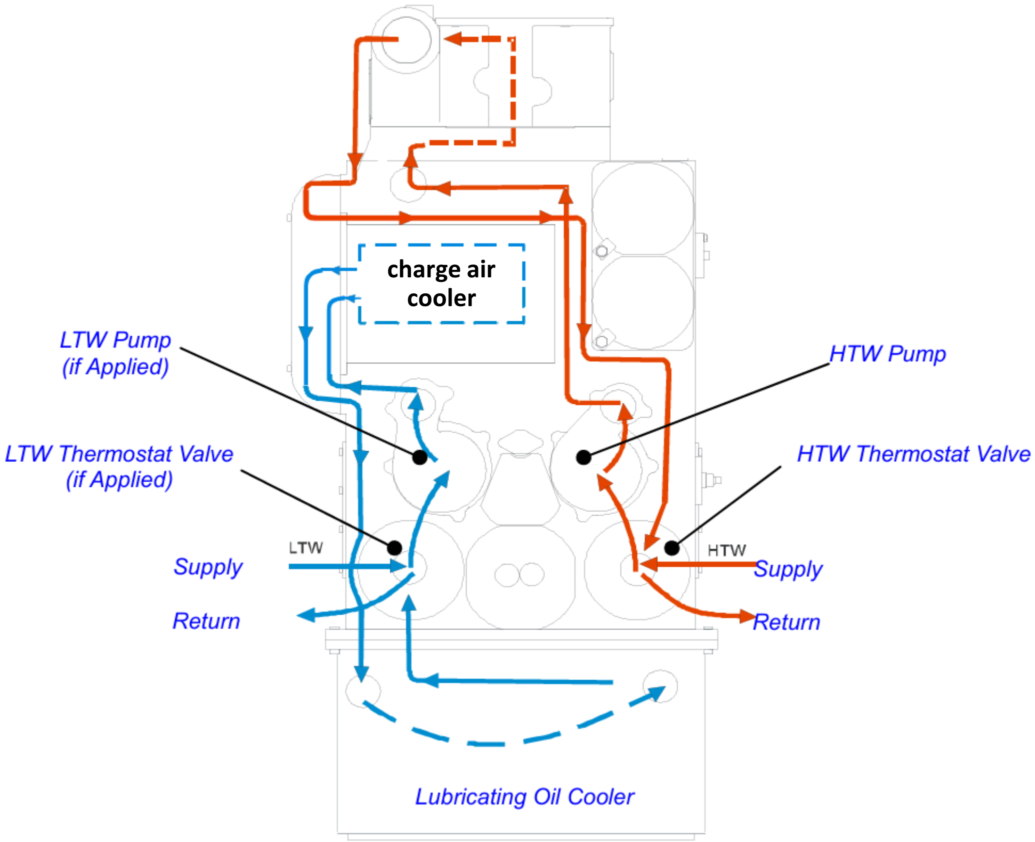
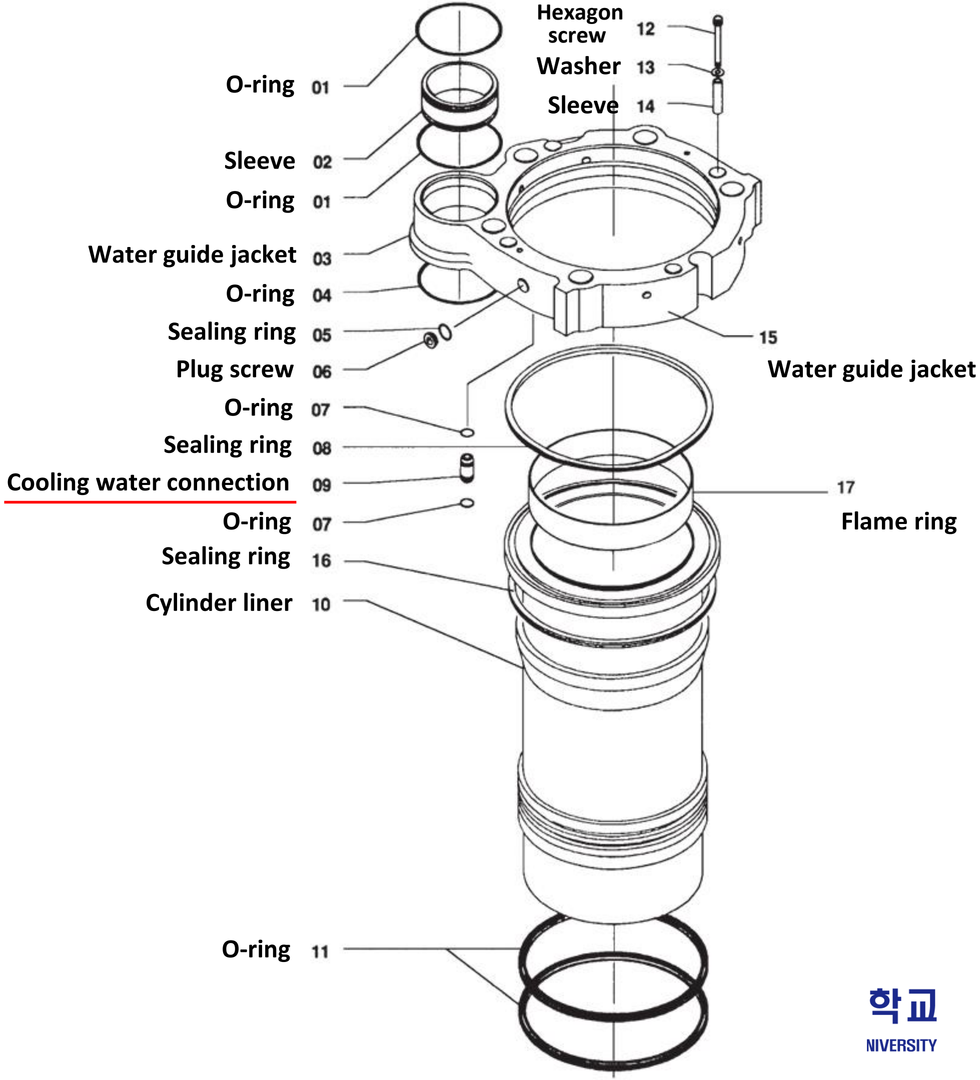


Fig.1 Circuit diagram for internal cooling water system (내부 냉각수 계통도)


O-ring 01
Sleeve 02
O-ring 01
Water guide jacket 03
O-ring 04
Sealing ring 05
Plug screw 06
O-ring 07
Sealing ring 08
Cooling water connection 09
O-ring 07
Sealing ring 16
Cylinder liner 10
O-ring 11
Hexagon screw 12
Washer 13
Sleeve 14
15 Water guide jacket
17 Flame ring

Cooling water system (HiMSEN)

The two cooling water pumps are of a centrifugal type and are driven by the gear wheel at the free end of the engine.

These two pumps are identical, but one is for circulating high temperature cooling water and the other is for low temperature cooling water.

Sealing rings are provided to avoid leakage between water side and oil side in the pump housing.

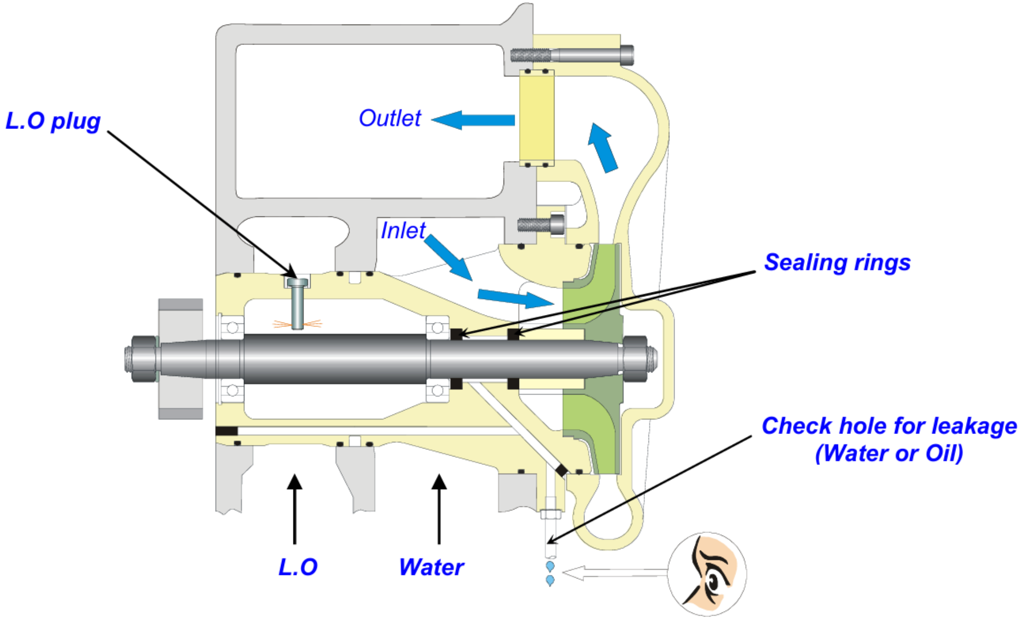
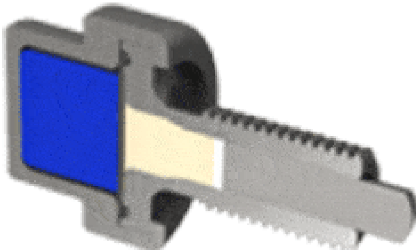


Fig.2 Check hole for leakage and cooling water flow in water pump
(점검 홀 및 펌프내의 냉각수 흐름)

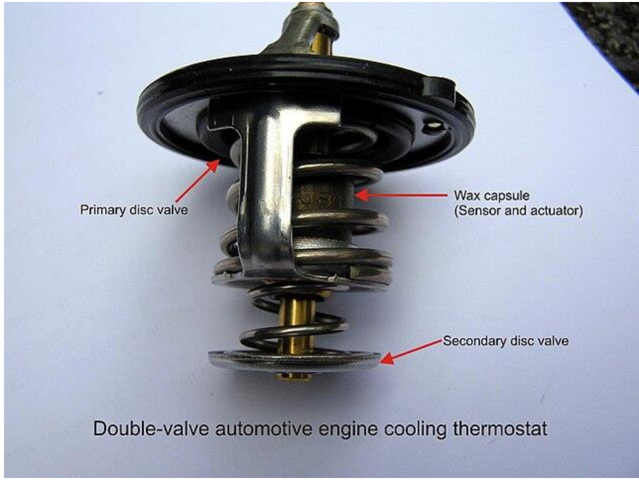
Cooling water system (HiMSEN)

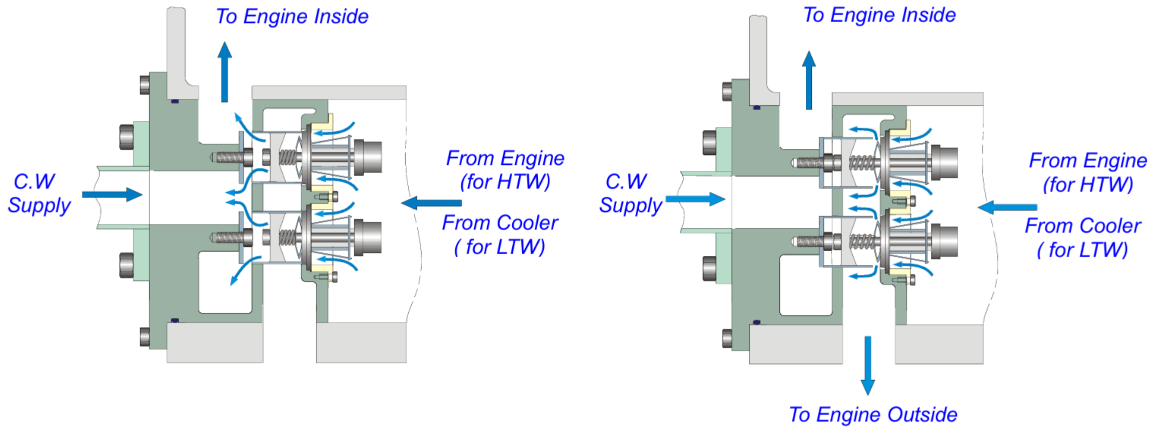
The two thermostat valves for HT and LT water circuits respectively are mounted on the feed block of the engine.

They are of the wax element type and cannot be adjusted. The thermostat valve controls the flow direction of the cooling water depending on the water temperature in the engine block appropriately as shown in fig. 3.



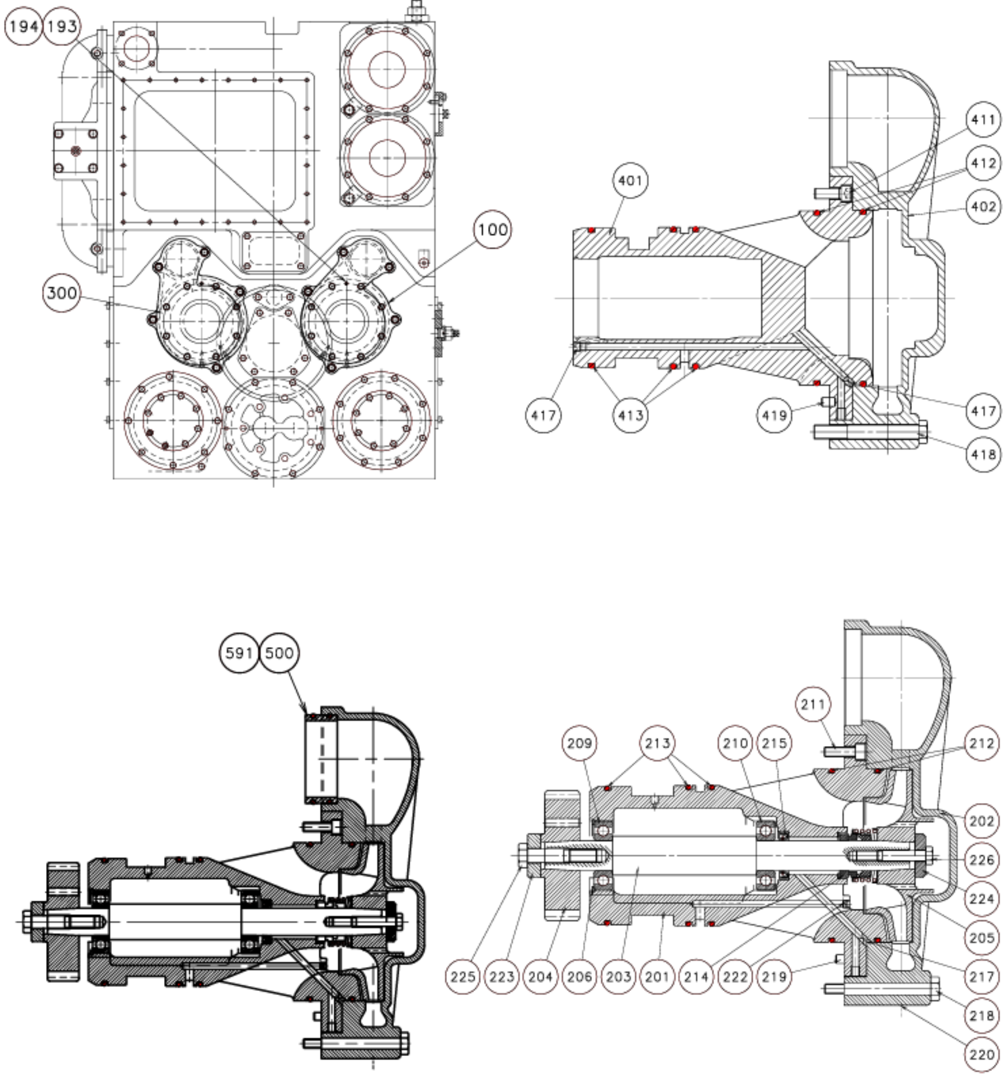
Wax in Solid State





(a) Low temperature cooling water *(b) High temperature cooling water*

Fig. 3 Cooling water flow in thermostat valve (온도 조정 밸브내의 냉각수 흐름)

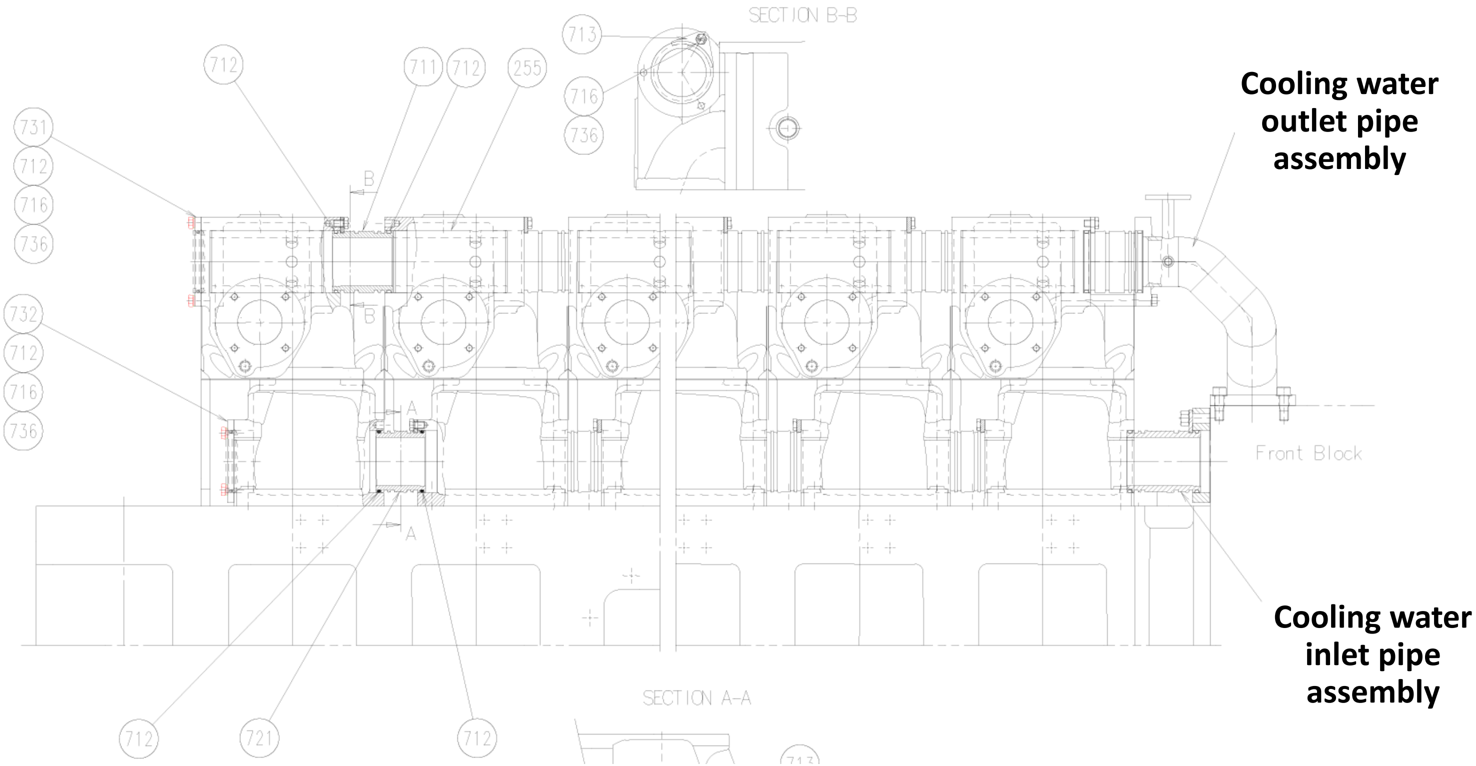


Part List

Item No.	Description	Q'ty / Eng.
100	H.T.Pump complete	1
193	Tab.dowel w.var.length	1
194	Nut	1
201	Pump casing	1
202	Pump cover	1
203	Shaft	1
204	Spur gear	1
205	Impeller	1
206	Retaining ring	1
209	Bearing	1
210	Bearing	1
211	Socket head bolt	8
212	O-ring	2
213	O-ring	3
214	Mechanical seal	1
215	Oil seal	1
217	Set screw	1
218	Bolt	5
219	Parallel pin	1
220	Plug screw	2
222	Plug	1
223	Washer	1
224	Washer	1
225	Bolt	1
226	Bolt	1
300	L.T.Blank	1
401	Pump casing	1
402	Pump cover	1
411	Socket head bolt	8
412	O-ring	2
413	O-ring	3
417	Set screw	2
418	Bolt	5
419	Parallel pin	1
500	Intermediate piece	2
591	O-ring	4

Cooling water system (HiMSEN)

Cooling water system (HiMSEN)



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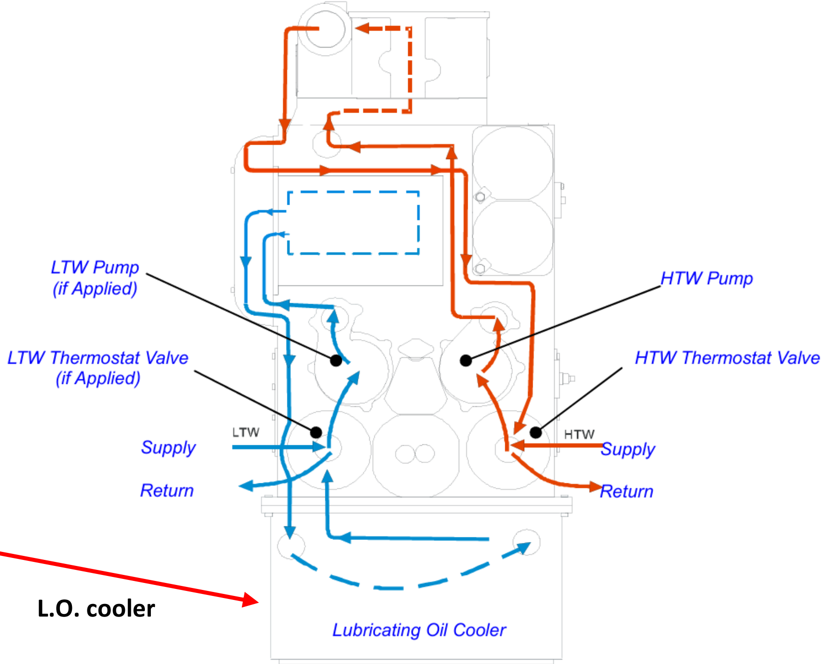

Fig.1 Circuit diagram for internal cooling water system (내부 냉각수 계통도)

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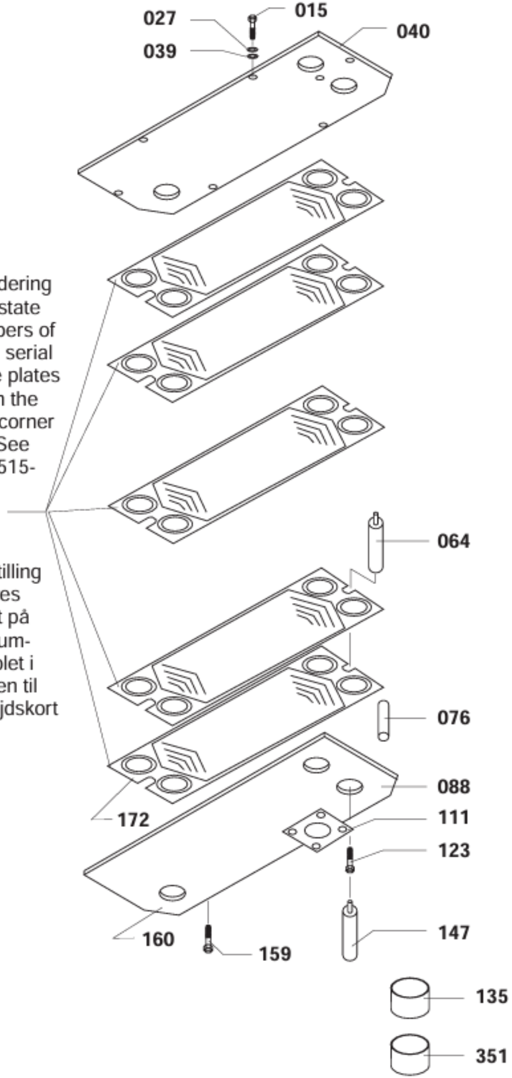
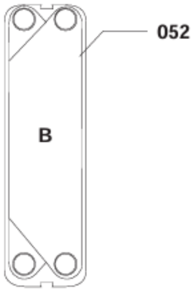
L.O. cooler

Note: When ordering plates, please state the serial numbers of the plates. The serial numbers of the plates can be found in the top right-hand corner of the plates. (See Working Card 515-06.00)

Husk: Ved bestilling af plader angives serienummeret på pladen. Serienummeret er stemplet i toppen af pladen til højre. (Se arbejdskort 515-06.00)





Item No.	Qty.	Designation
015	4/K	Hexagon screw
027	4/K	Washer
039	4/K	Washer
040	1/K	Pressure plate
052	/I	Gasket
064	2/K	Guide bar
076	4/K	Distance piece
088	1/K	Frame plate
111	4/K	Gasket
123	2/K	Screw
135	1/K	Glue
147	2/K	Guide bar (for dismantling)
159	32/K	Screw
160	1/E	Lubricating oil cooler, complete
172	/I	Plates
351	/I	Cleaning fluid

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Plate type heat exchanger (=Cooler)

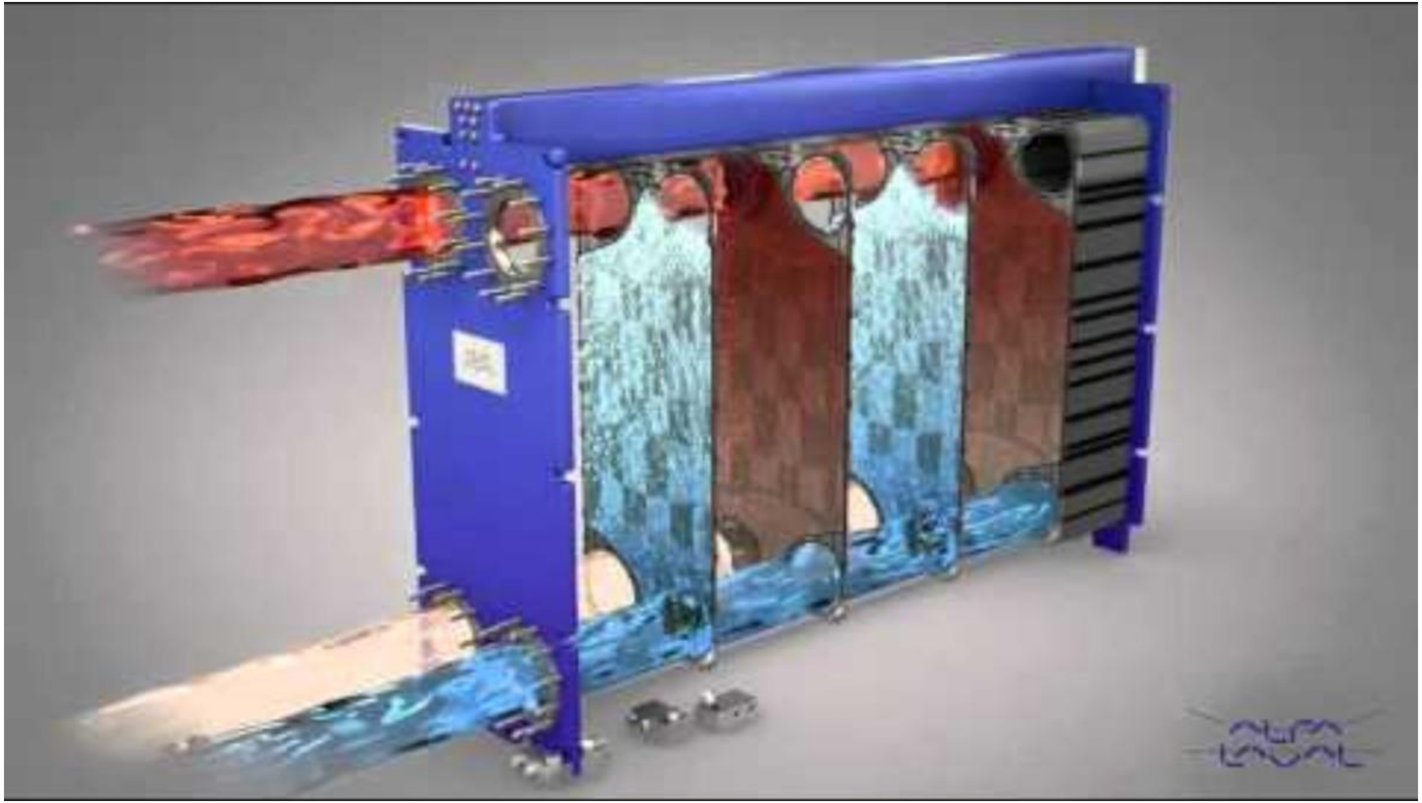

Plate type heat exchanger (=Cooler)

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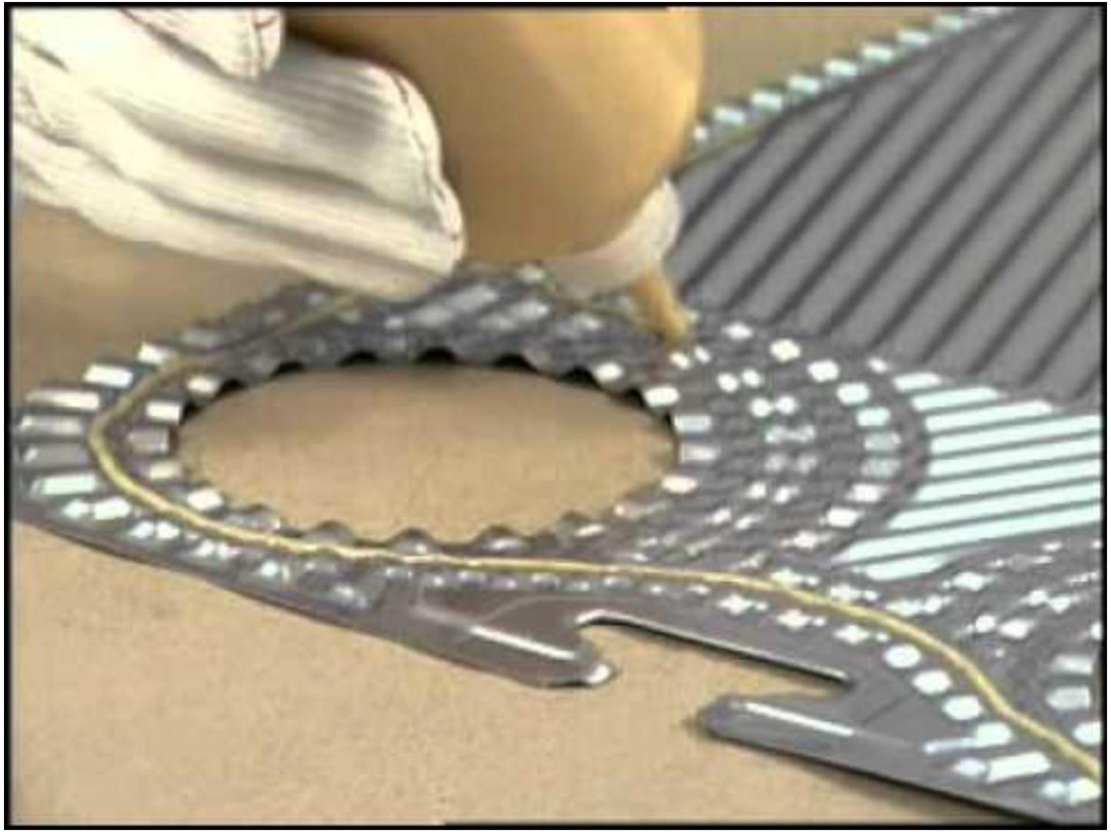

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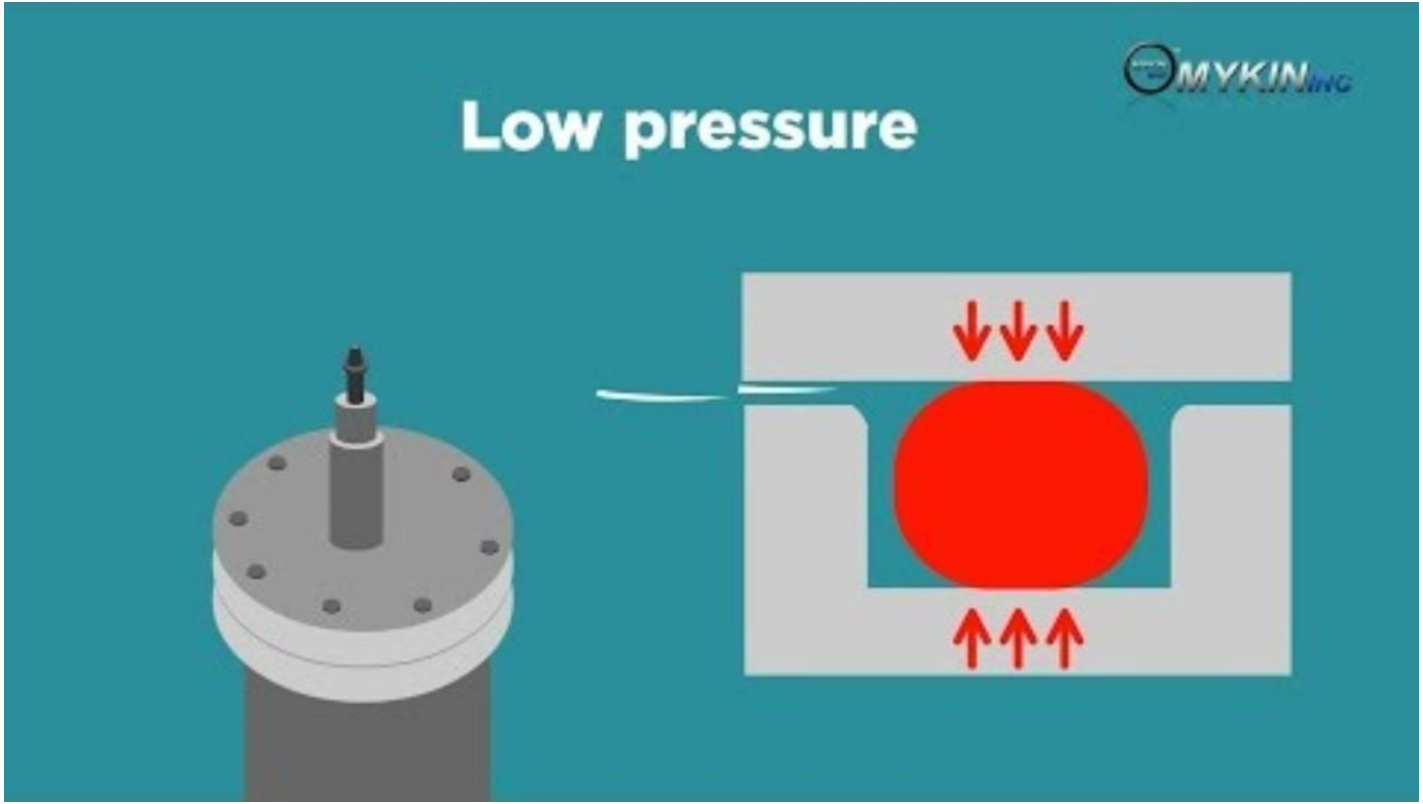
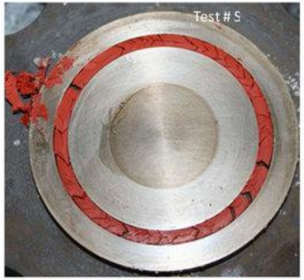


Plate type heat exchanger (=Cooler)

In plate type heat exchangers, a rubber gasket is used to separate the two fluid flows.

However, rubber gaskets harden when exposed to heat for a long period of time, becoming less elastic and thus hindering the separation of the two fluids.

In particular, when the plate inside the plate type heat exchanger is disassembled for cleaning, the elasticity and damage of the gasket should be checked, and any problematic gaskets should be replaced with spare gaskets.



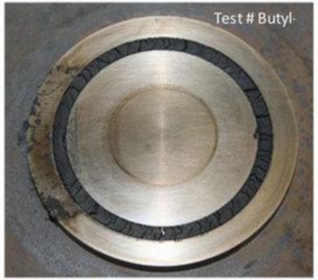
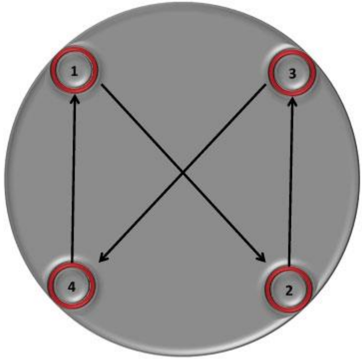


Plate type heat exchanger (=Cooler)



After cleaning the plate, when assembling the plate exchanger, tighten the tightening nut in an X shape to ensure that the rubber gasket is evenly sealed throughout.

If you tighten one nut as tight as possible and then tighten the other nut as tight as possible, the rubber gasket will not seal evenly, causing leakage.

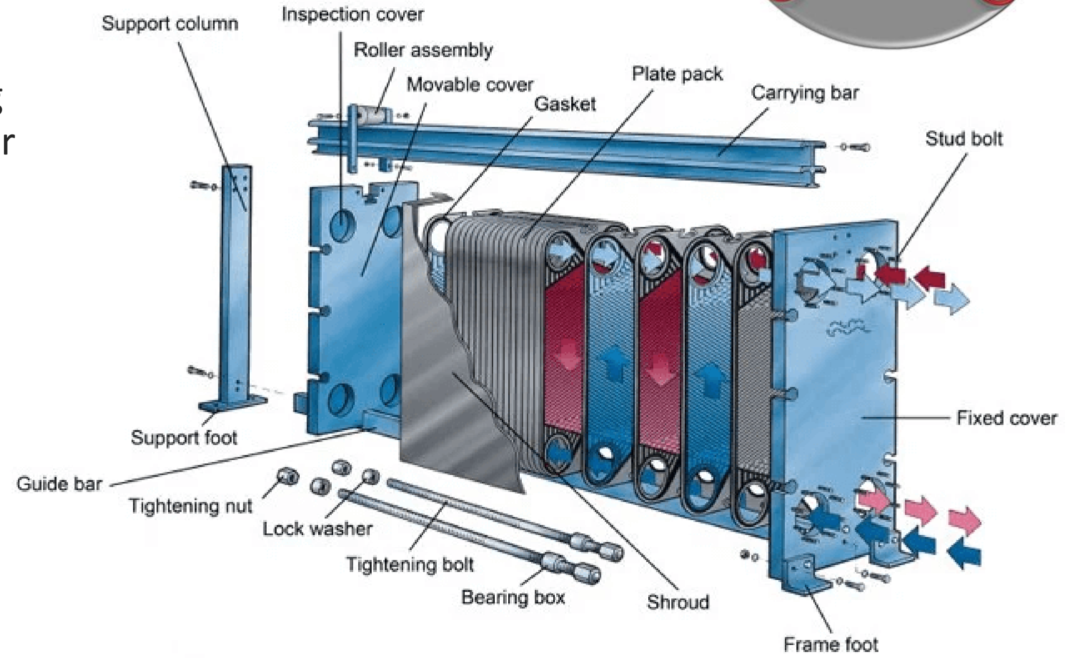


Plate type heat exchanger (=Cooler)

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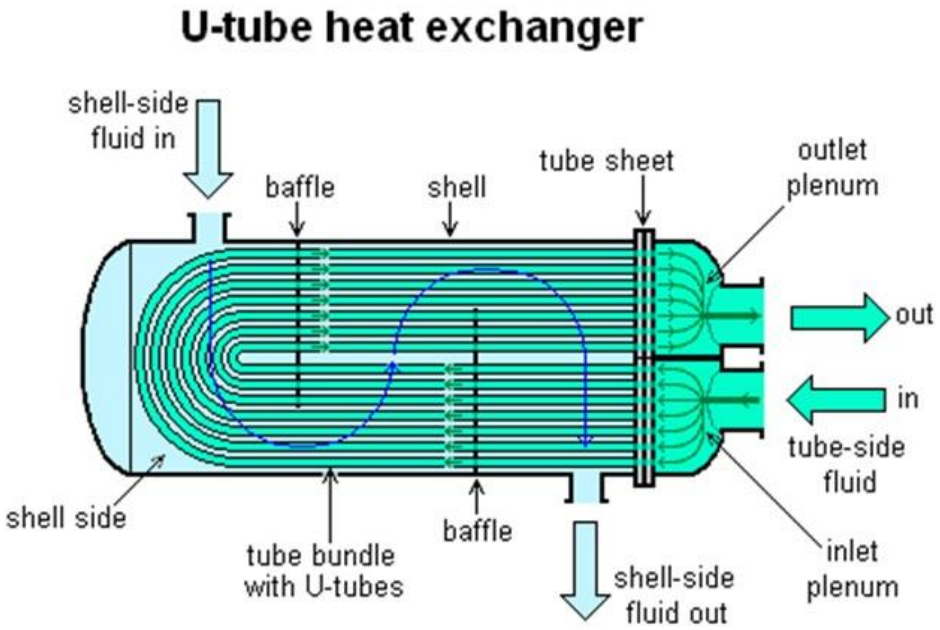
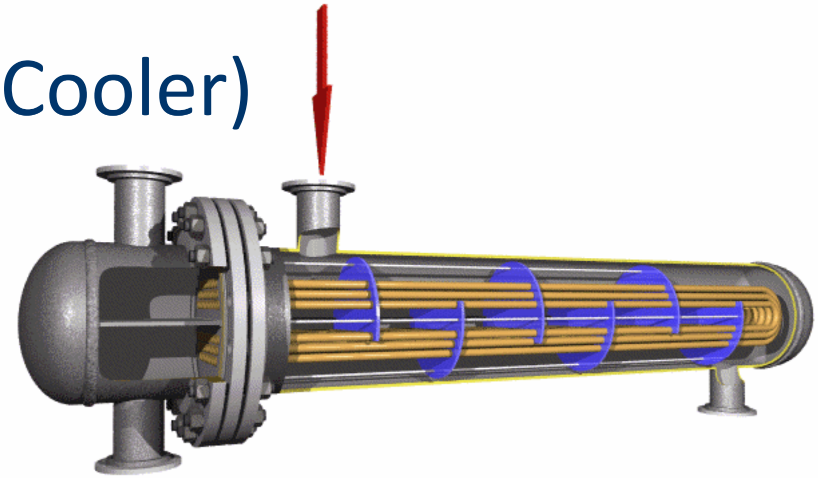
Plate type heat exchanger (=Cooler)

Shell type heat exchanger (=Cooler)

The L.O. cooler of some generators uses a shell type heat exchanger.





Air Cooler (HiMSEN)

Charge air cooler

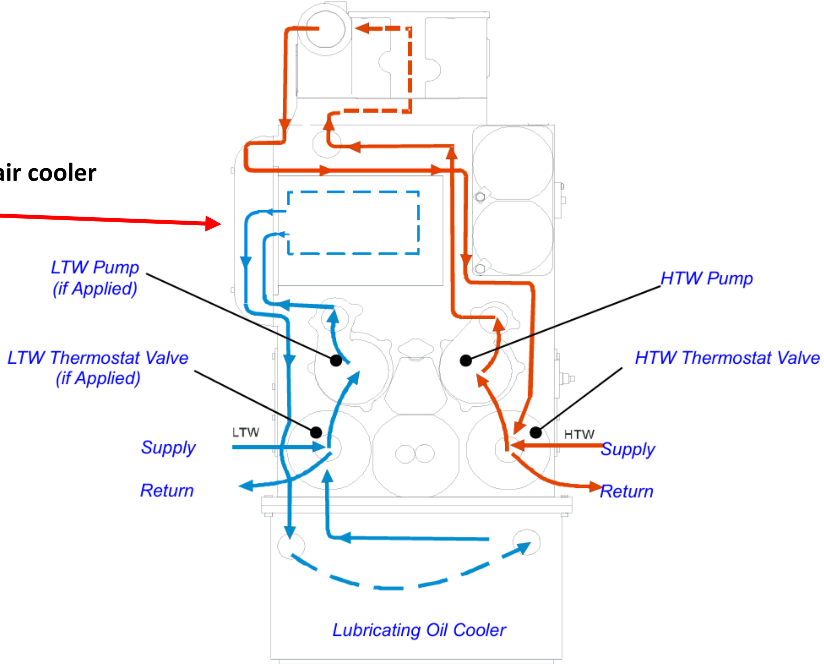
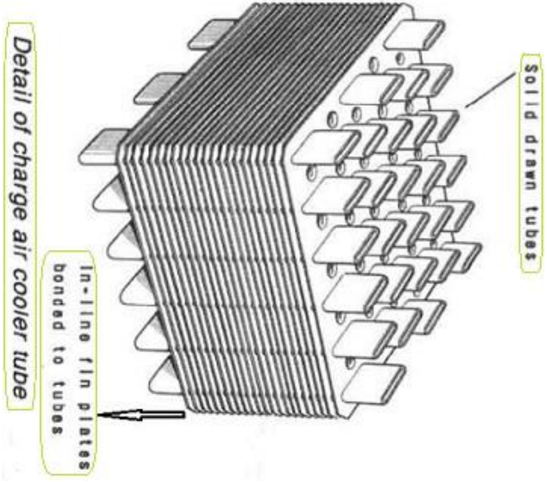


Fig.1 Circuit diagram for internal cooling water system (내부 냉각수 계통도)

Air Cooler (HiMSEN)

Cooling water flows through the tubes and charged air from the turbocharger flows between the fins.

Accordingly, heat exchange occurs between the cooling water and the charged air, so that the charged air is cooled and supplied into the cylinder.



Air Cooler (HiMSEN)

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Cooling water treatment test

1) Nitrite test, 2) Chloride test, 3) pH test

A cooling water treatment test is conducted to determine the current condition of the G/E's cooling water.

Based on the test results, chemicals are added to the cooling fresh water expansion tank to prevent corrosion and suppress scale caused by cooling water.

References

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6) https://www.youtube.com/watch?v=s5nJG0om8KY

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